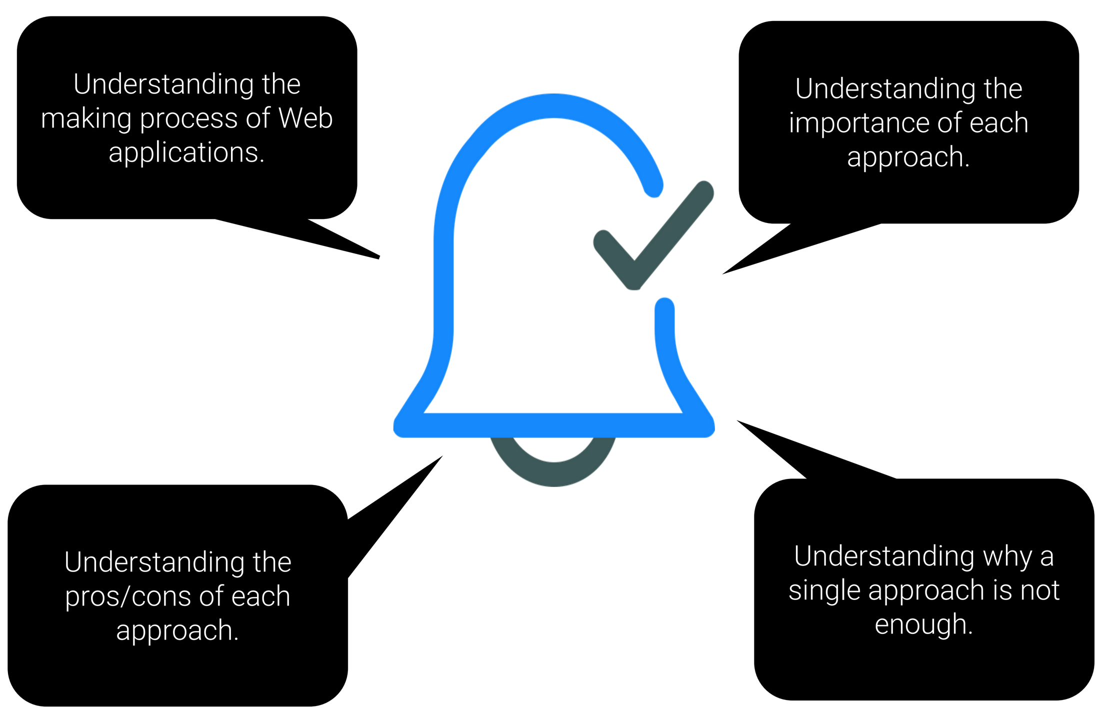
Comparing Static-Site Generation to Single-Page Application and Server-Side Rendering as approaches to modern Web applications.

COMMUNICATION AND MEDIA ENGINEERING (CME) MASTER THESIS, SUMMER SEMISTER 2022

Abstract

- This master thesis deals with the three most popular ways of making Web applications today. Server-Side Rendering (SSR), Single-Page Application (SPA), and Static-Site Generation (SSG).
- The main point of this thesis is to get familiar with the concept of modern web development not only from front-end to back-end but also different methods of creating modern full stack web applications.
- For implementation same Web application has been developed in three different ways and compared based on flexibility, performance, data fetching, security, learning curve, popularity, and SEO.

Goal



Overview

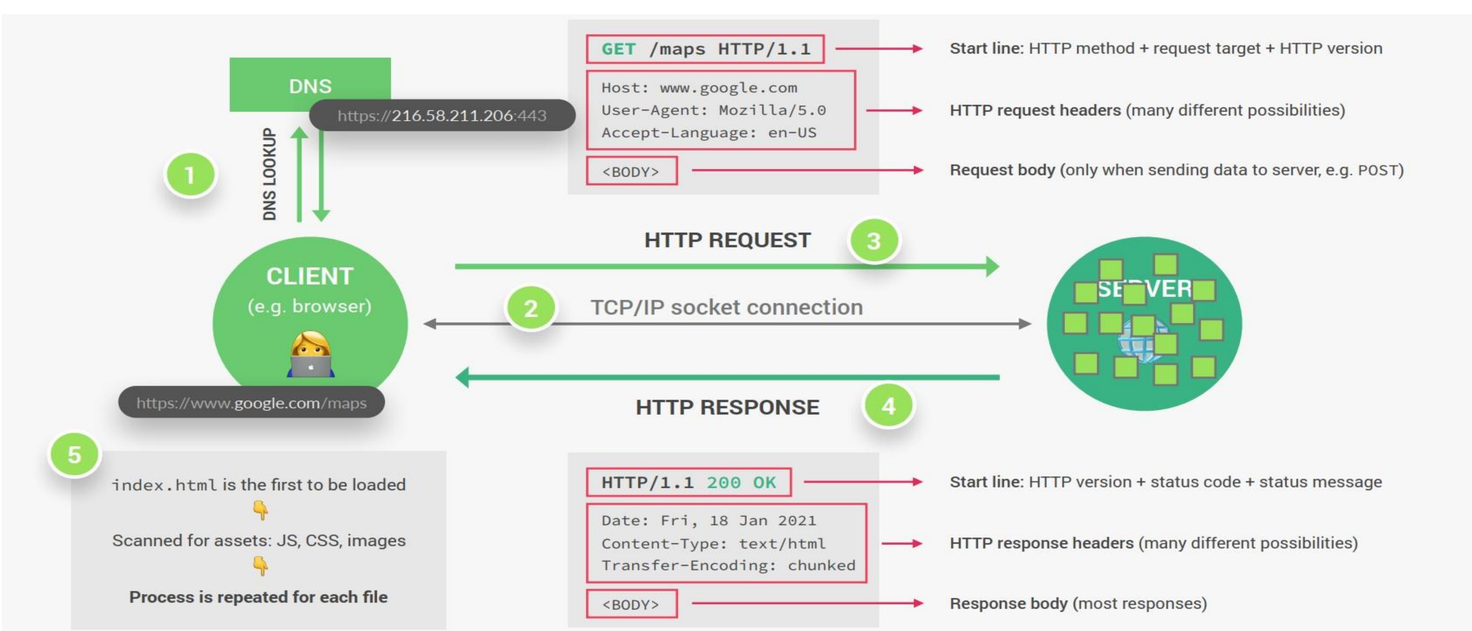


Figure 1: Deep inside how we get access to the web

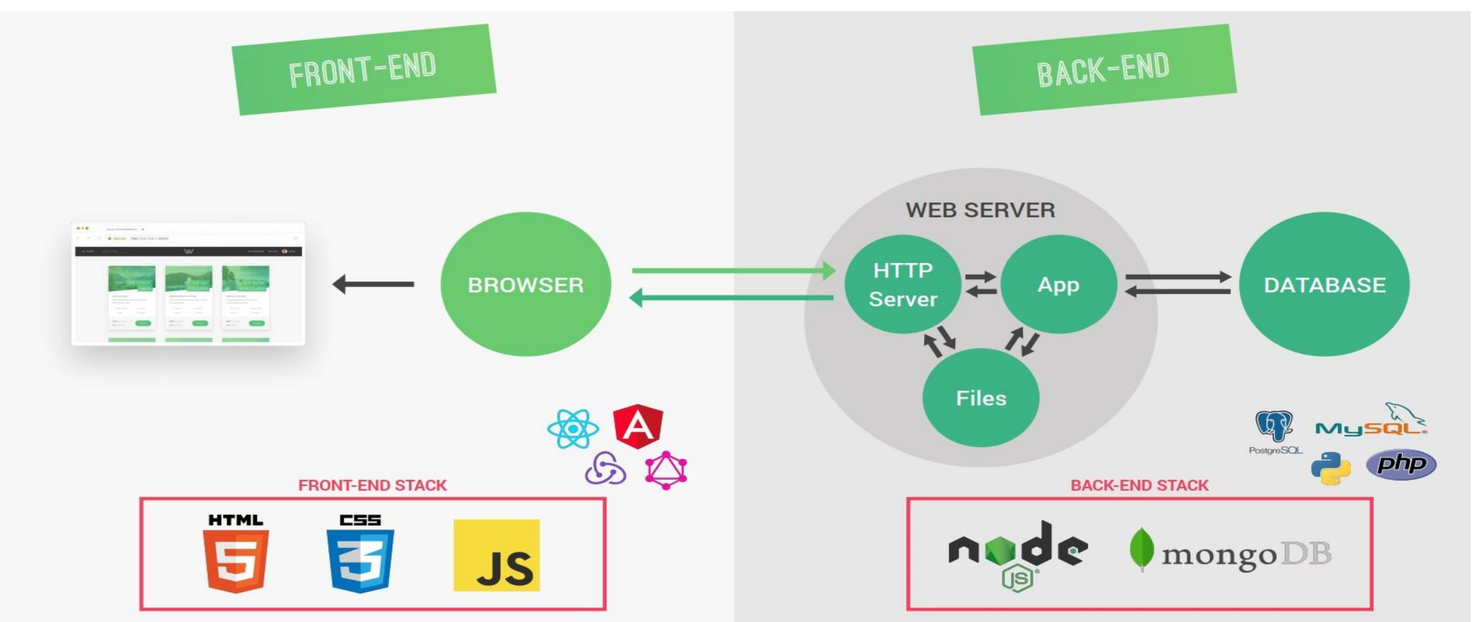


Figure 2: Front-end and back-end in Web development and the technologies.

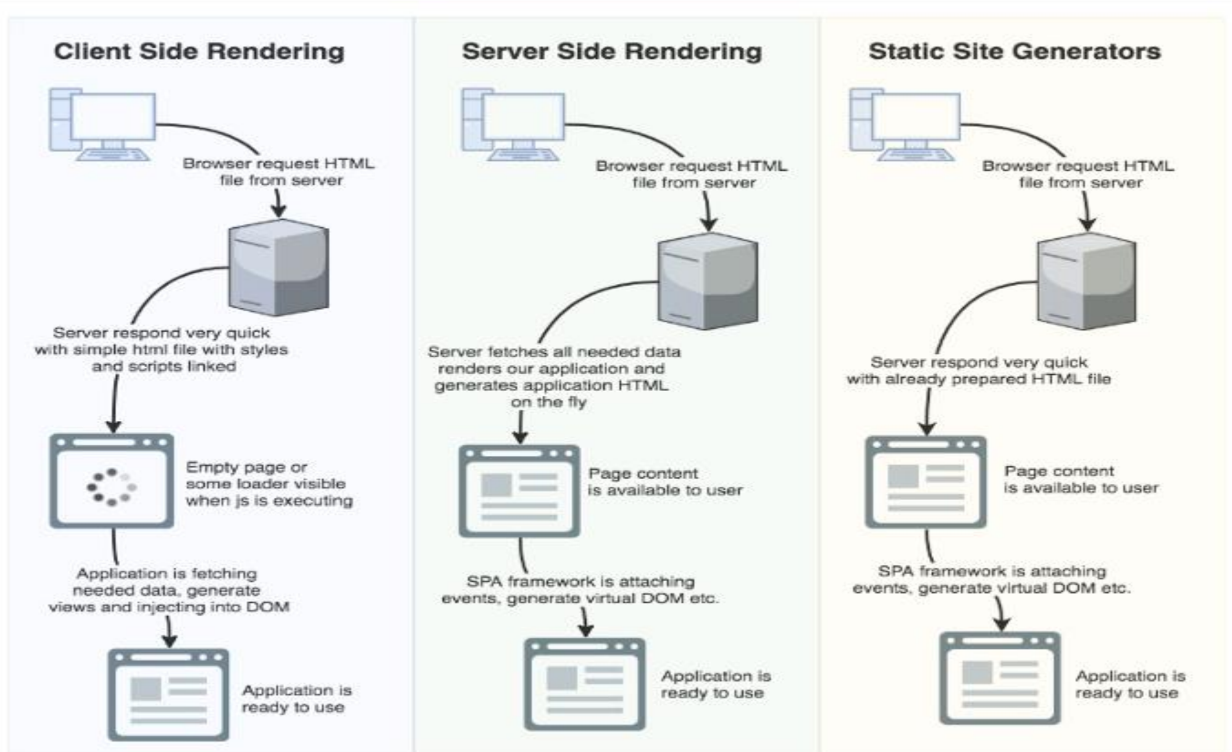
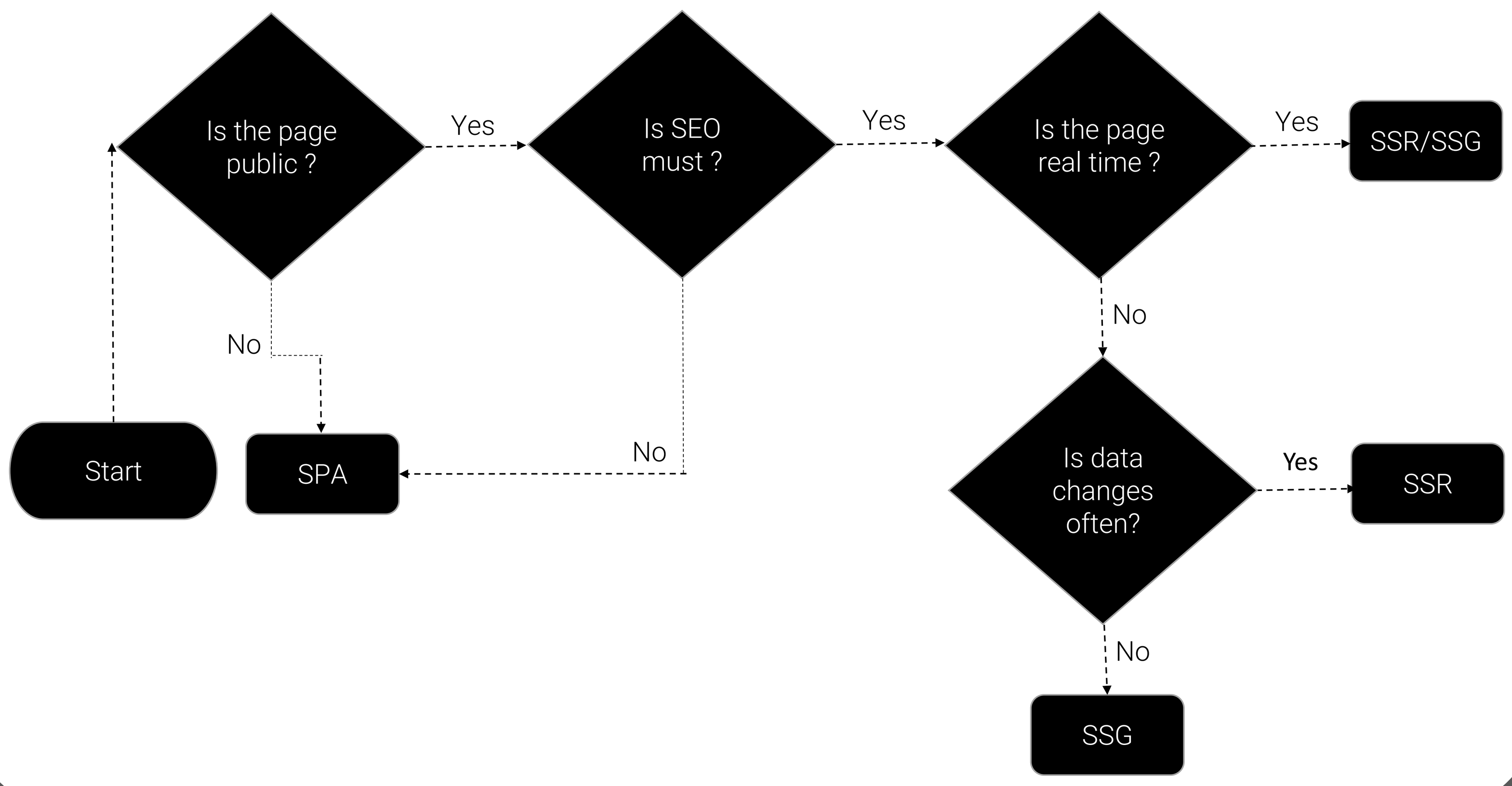


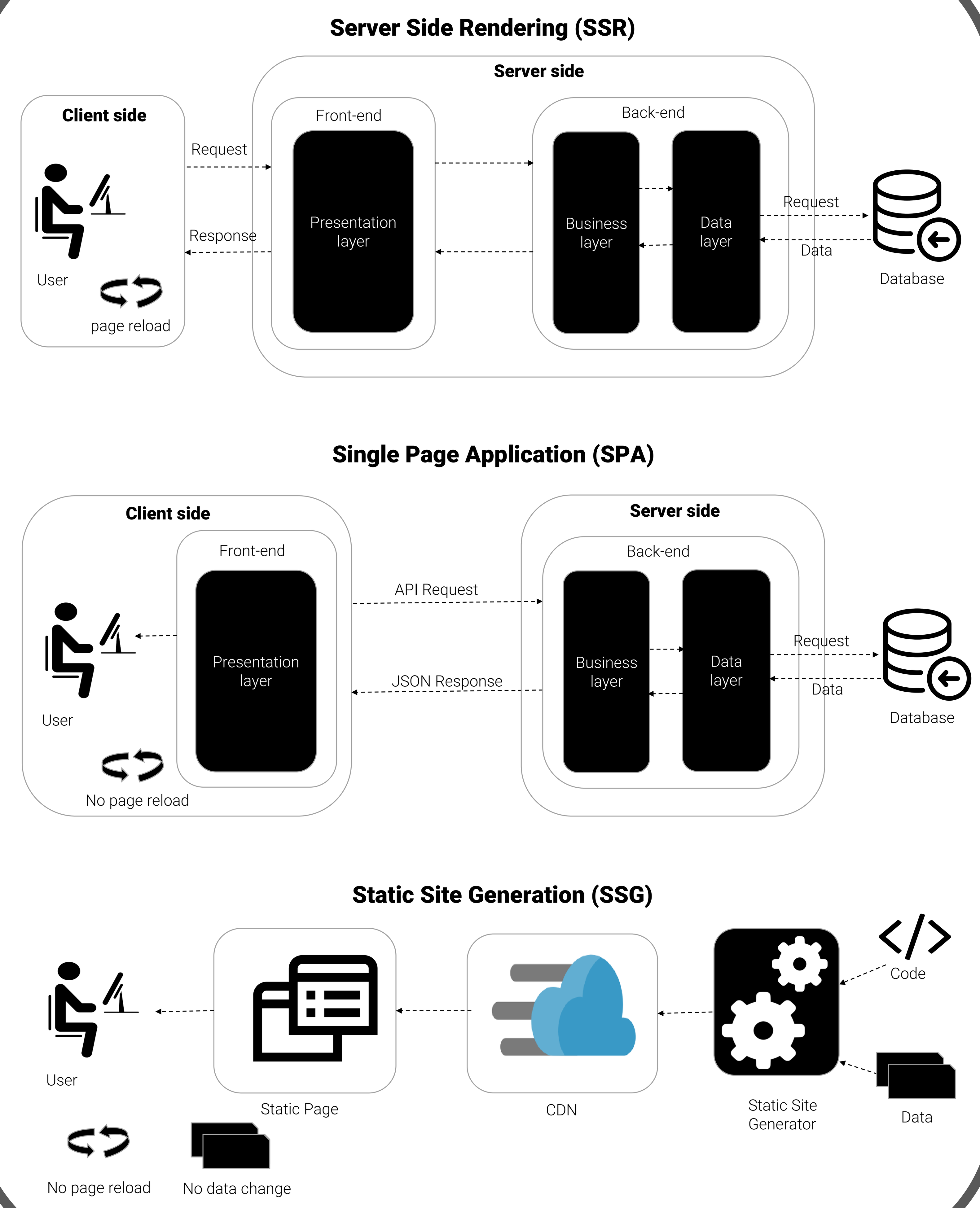
Figure 3: Three different approaches of making Web applications today.

Flowchart



A flowchart to determine when to use SSR, SPA, and SSG

Architecture



References

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2. Figure 3 : https://medium.com/verclaire-nine/client-side-rendering-vs-server-side-rendering-vs-static-site-generation.
3. https://mobidev.biz/blog/web-application-architecture-types
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